

EOM231.03

DESCRIBE THE TYPES  
OF DRAG THAT ACT  
UPON AN AIRCRAFT

# Teaching Points

TP1 Introduce Drag as a Useful Force and Explain the Two Types of Drag

TP2 Define the Components of Parasite Drag

TP3 Demonstrate Form Drag

TP4 Explain Induced Drag

TP5 Demonstrate Induced Drag

## TP1 Introduce Drag as a Useful Force and Explain the Two Types of Drag

- Drag is the force that **opposes the forward motion** of an aircraft.

### Parasite Drag

- Caused by those parts of the aircraft that **DO NOT** generate lift
- Includes drag caused by openings, such as those in the cowlings and those between the wing and the ailerons and flaps

### Induced Drag

- Caused by those parts of the aircraft that **DO** generate lift (wings)
- Induced drag is the result of the wing and is therefore a part of lift and **can never be eliminated**

# TP1 Introduce Drag as a Useful Force and Explain the Two Types of Drag

## DRAG DEFINITIONS

**DRAG** IS THE FORCE THAT OPPOSES THE FORWARD MOTION OF AN AIRCRAFT.

- THE TWO MAIN TYPES OF DRAG ARE **PARASITE DRAG** AND **INDUCED DRAG**.
- PARASITE DRAG IS FURTHER BROKEN INTO TWO TYPES, **FORM DRAG** AND **SKIN FRICTION**.

Which of the following will not minimize parasite drag if done

- A) Having retractable landing gear
- B) Tightening bolts and rivets
- C) Reducing wingspan
- D) Cleaning the aircraft
- E) None of the above

## TP2 Define the Components of Parasite Drag

- Parasite drag is broken down into two components:
- **Form drag** refers to the **drag created by the form or shape** of a body as it resists motion through the air
- **Skin friction** refers to the **tendency of air flowing over a body to cling to its surface**
- Although parasite drag can never be eliminated, it can be reduced. One method is to **remove parts of the aircraft** that cause it. Another method is to **streamline those parts** that cannot be eliminated. Skin friction can be reduced substantially by the **removal of dust, dirt, mud or ice** that has collected on the aircraft



# **BONUS VIDEO:**

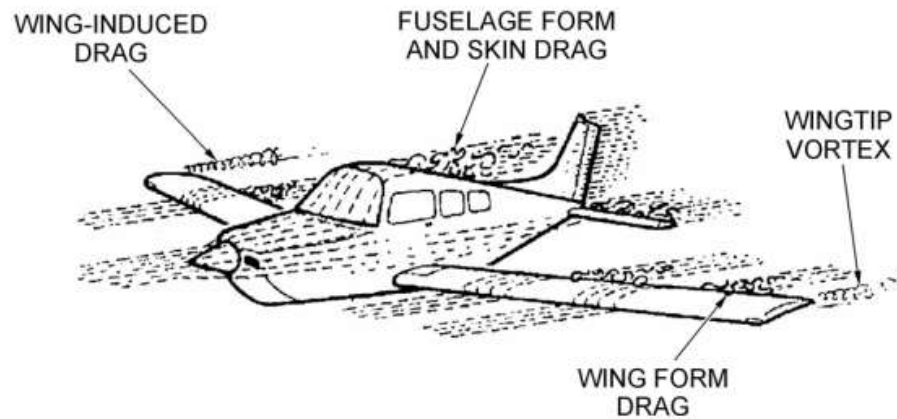
## Parasite Drag

Form drag is caused by \_\_\_\_\_ while skin friction is caused by \_\_\_\_\_.

- A) weight, air
- B) design, shape
- C) shape, air
- D) weight, shape
- E) two of the above are correct



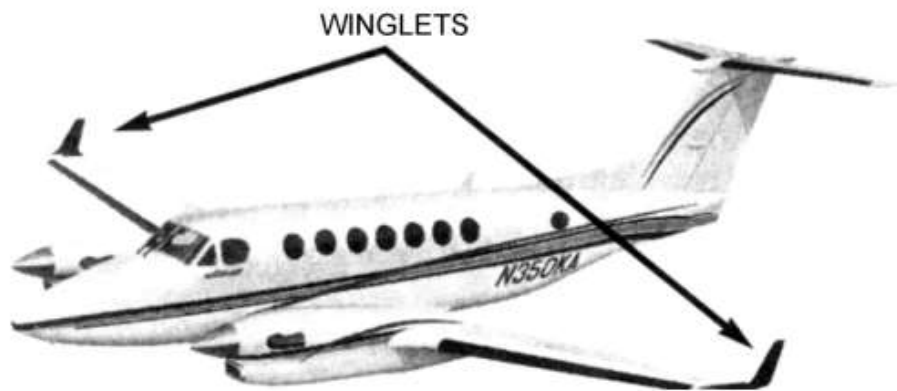
## TYPES OF DRAG



Cadets Canada: RCSU Pacific, 2007, Air Cadet Master Lesson Plans. Retrieved 7 March 2007, from [http://www.regions.cadets.ca/pac/aircad/resources/mlp\\_air\\_e.asp](http://www.regions.cadets.ca/pac/aircad/resources/mlp_air_e.asp)

Figure D-1 Types of Drag on an Aircraft

## PREVENTING WING TIP VORTICES



A-CR-CCP-263/PT-001 (p. 27)

Figure D-2 Winglets

# TP4 Explain Induced Drag

- Induced drag is produced by those **parts of an aircraft that are active in producing lift**. Induced drag results from the wing and is therefore a part of lift that can never be eliminated. Induced drag can only be **reduced during the initial designing of the aircraft**.
- Induced drag increases as the **angle of attack increases** and decreases as the AoA decreases.



# What law or principle is the cause of wingtip vortices?

- A) Newton's First law
- B) Newton's Second Law
- C) Newton's Third Law
- D) Bernoulli's Principle
- E) Air's property as a fluid